

PARAKH — Stage 4 PRD

Risk Signal Computation Layer

1. Objective

Compute **raw anomaly signals** for each record that capture statistically unusual behavior **independent of confidence**.

Each record must produce a feature vector:

\$\$
 $X(r) = \{z_{\{\text{cost}\}}, \text{HHI}, B, D_{\{\max\}}\}$
\$\$

These are **not final risk scores**—they are interpretable signals used in Stage 5.

2. Scope

Included

- Cost outlier detection
- Vendor concentration (HHI)
- Temporal burst detection
- Duplicate signal integration
- Feature normalization (pre-aggregation)

Excluded

- Final risk scoring
 - Confidence computation
 - Routing decisions
-

3. Inputs

From Stage 1:

```
Corpus.records
```

From Stage 3:

```
SemanticResult:  
- cluster_id
```

- cost_stratum
- peer_cell
- duplicate_score (D_max)

4. Outputs

Each record must produce:

```
{
  "z_cost": float,
  "hhi": float,
  "burst_score": float,
  "duplicate_score": float
}
```

5. Cost Outlier (Primary Signal)

5.1 Definition

$$z_{\text{cost}}(r) = \frac{\log a_r - \operatorname{med}_{\{k,s\}}(\log a)}{1.4826 \cdot \operatorname{MAD}_{\{k,s\}}(\log a)}$$

Where:

- (k,s): peer cell
- (a_r): sanction_amount

5.2 Implementation Steps

1. Group records by peer cell
2. Compute:
 - median(log cost)
 - MAD(log cost)
3. Compute z-score per record

5.3 Edge Handling

If:

- `MAD < 1e-6` OR
- `cell_size < n_min`

Then:

```
z_cost = None
```

5.4 Output Range

- Normal: ~[-3, +3]
- Extreme outliers: > |5|

6. Vendor Concentration (HHI)

6.1 Definition

For each cell ((c,k)):

$$\text{HHI}_{c,k} = \sum_v \left(\frac{a_v}{\sum a} \right)^2$$

Where:

- (a_v): total awarded to vendor v

6.2 Implementation

1. Group by:

```
(constituency/district, cluster_id)
```

2. Aggregate:

- total spend per vendor
- total spend in cell

3. Compute HHI

6.3 Assignment

Each record inherits:

```
hhi = HHI(cell)
```

6.4 Edge Cases

- Single vendor → $HHI = 1$
 - Missing vendor → exclude or assign neutral value
-

7. Temporal Burst Detection

7.1 Definition

\$\$

$B(d,m) =$

$\log \frac{n_{\{d,m\}} + \alpha}{\lambda_d(m) + \alpha}$

\$\$

Where:

- $(n_{\{d,m\}})$: records in district d , month m
 - $(\lambda_d(m))$: seasonal baseline
-

7.2 Baseline Estimation

For each district:

- Compute monthly averages across years
 - Use **month-specific baseline** (March vs March)
-

7.3 Parameters

```
alpha = 1.0
```

7.4 Assignment

Each record gets:

```
burst_score = B(district, month(date_approval))
```

7.5 Interpretation

- 0 → higher-than-expected activity
 - < 0 → lower-than-expected
-

8. Duplicate Signal

Already computed in Stage 3:

```
duplicate_score = D_max
```

8.1 Validation

Ensure:

- identical names → high score (~0.9+)
 - unrelated → near 0
-

9. Feature Normalization

Before Stage 5, normalize all signals:

9.1 Method

```
percentile_rank(feature)
```

9.2 Rules

- Apply globally (NOT mixed scopes)
 - Ignore None values
 - Clip extreme values
-

9.3 Output

```
{  
  "z_cost_norm": [0,1],
```

```
"hhi_norm": [0,1],  
"burst_norm": [0,1],  
"dup_norm": [0,1]  
}
```

10. Implementation Architecture

10.1 Modules

- `cost_outlier.py`
- `hhi.py`
- `burst.py`
- `normalization.py`

10.2 Core Class

```
class RiskSignals:  
    def compute(self, corpus, semantic_result):  
        return RiskFeatureResult
```

10.3 Output Object

```
class RiskFeatureResult:  
    raw_features: Dict  
    normalized_features: Dict
```

11. Non-Functional Requirements

Performance

- 50k records processed in < 10 seconds

Stability

- No NaN/inf values

Determinism

- Same input → same output
-

12. Validation & Testing

12.1 Cost Outlier Checks

- median-cost records $\rightarrow z \approx 0$
 - extreme cost $\rightarrow$ large $|z|$
-

12.2 HHI Checks

- uniform vendors $\rightarrow$ low HHI
 - dominant vendor $\rightarrow$ high HHI
-

12.3 Burst Checks

- uniform distribution $\rightarrow$ near 0
 - spikes $\rightarrow$ positive
-

12.4 Duplicate Checks

- identical entries $\rightarrow$ high score
 - random entries $\rightarrow$ low
-

13. Edge Cases (MANDATORY)

Handle:

- Missing cost values
 - Single-record peer cell
 - No vendor info
 - All records in same month
 - Zero variance in cost
-

14. Acceptance Criteria

Stage complete if:

- ☐ All signals computed per record
 - ☐ Invalid cases handled safely
 - ☐ Normalized features in $[0,1]$
 - ☐ No crashes on edge cases
 - ☐ Works on synthetic dataset
-

15. Definition of Done

- All anomaly signals computed
 - Ready for aggregation (Stage 5)
-

16. Key Design Principle

Signals, not decisions.

This stage does not decide fraud.

It produces **independent, interpretable indicators** of unusual behavior.
